Adventist University of Central Africa

MSc in Big Data analytics

Data Mining

Lab: 2

Task Context: You will work individually, prepare a dataset for analysis, perform exploratory data analysis and feature engineering, perform hyperparameter tuning, and build and evaluate machine learning models.

Task Objectives:

a) Perform data preprocessing
b) Perform feature engineering
c) Perform hyperparameter tuning
d) Build and evaluate ML models

Toolkit:

- numpy, pandas, and matplotlib, sklearn, etc.

Task Description:

Dataset: Consider the credit approval dataset available here.

Required

Note: You should include Markdown cells and comments to describe the tasks you are performing. Include comments to explain your codes.

1) *Data Preparation*: Download the dataset, load it into a pandas data frame, and prepare it for analysis. Write the resulting pandas data frame to a csv file
2) *Exploratory Data Analysis (EDA)*: Use numpy, pandas, and matplotlib to perform EDA on the data. This should include identifying the presence of missing values (if any), creating visualizations with the data, and identifying presence of outliers (if any).
3) *Preprocessing, Feature Selection and Engineering*: This should make your data ready for building machine learning models. Using pandas and scikit-learn: a) Handle missing values and outliers (if any), and

b) Perform appropriate label encoding (for categorical attributes) and data scaling (if needed).

4) *Model creation and evaluation*:

a) Using scikit-learn, build a classification-based model. Provide results using two performance metrics (of your choice). Justify the choice of metrics.

b) Using scikit-learn, build a neural network-based model and perform evaluation using 10-fold cross validation. Your model should be multi-layer. Provide results using two performance metrics (of your choice). Justify the choice of metrics.

c) Using hyperparameter tuning (with Grid Search), and based on 10-fold crossvalidation, identify the combination of hyperparameters that gives the best estimator on the dataset. Your search space for hyperparameters must include at least three hyperparameters - selected from learning rate, epochs, activation function, optimizer, batch-size, among others.

d) Compare the performance of the tuned model in b) against that in the model in a). Use visualizations to highlight the comparison.